Example 1

```
public class A_ArrrayDemo
{
    public static void main(String args[])
    {
         int a[]=new int[5];//declaration and
instantiation
         a[0]=10;//initialization
         a[1]=20;
         a[2]=70;
         a[3]=40;
         a[4]=50;
         System.out.println(a[6]);

        /*
        String name = null;

         System.out.println(name.length());
        */
}
}
```

Example 2- user defined exception

```
// User-defined exception
class InvalidAgeException extends Exception {

     /*
      public Exception(String message) {
        super(message);//Throwable
      }


       public Throwable(String message) {
         detailMessage = message;
    }


      */
```

```
     //  throw new InvalidAgeException("Age must be 18 or
above");
    InvalidAgeException(String message) {
        super(message);//Exception class

    }
}

public class AgeExceptionMain {

    static void checkAge(int age)
    {

     try
     {
           if (age < 18) //15<18
           {
                 throw new InvalidAgeException("Age must
be 18 or above");
              }

              System.out.println("Eligible to vote");
     }

     catch(InvalidAgeException i)
     {
          System.out.println(i);
     }
    }

    public static void main(String[] args) {


          checkAge(15);

    }
}
```

Example 3

```
public class ArrayDemo
{
    public static void main(String args[])
    {
     int a = 5;
          int b = 0;
          int result = a/b;
          System.out.println(result);

}
}
```

Example 4

```
/*
class B_ExceptionIntro
{
     public static void main(String args[])
     {
          int a = 5;
          int b = 0;
          int result = a/b;
          System.out.println(result);
     }
}
*/

public class B_ExceptionIntro {
    public static void main(String args[])
     {
          int a = 5;
          int b = 0;

          try
          {
               int result = a/b;
               System.out.println(result);
          }
```

```
          catch(ArithmeticException exception)
          {
            System.out.println("Can not divide by
zero.Denominator values is 0.Change denominator value to
non zero value");
            //System.out.println(exception);
          }
     }
}
```

Example 5

```
class C_BasicProgram
{
  public static void main(String args[])
  {
    try
    {
          System.out.println("Before Division");
          int i = 1;
          int j = 0;
          System.out.println(i/j);
          System.out.println("After Division");

       /*
        int number = Integer.parseInt("a");
        String name = null;
        System.out.println(name.length());
        */


          /*
          String s = "123";
          int n = Integer.parseInt(s);


          String s = "abc";
          int n = Integer.parseInt(s);  //
NumberFormatException
```

```
          */

    }



    catch(ArithmeticException e)
    {
      System.out.println("ArithmeticException - Dr can not
be zero");
    }


    catch(ArrayIndexOutOfBoundsException e)
    {
      System.out.println("ArrayIndexOutOfBoundsException -
check the array values");
    }

    catch(NumberFormatException e)
    {
      System.out.println("ArrayIndex OutOfBoundsException
= String does not contains valid number");
    }

    catch(Exception exception)
     {
          System.out.println(exception);
     }


     finally
    {
      System.out.println("Finally block");
    }

  }
}
```

Example 6

```
public class D_ThrowDemo {
    static float validate(int a, int b)
    {
          if(b==0)
          {
              throw new ArithmeticException("Cannot
                          divide by zero ");
          }
          else
          {
            float  result = a/b;
             return result;
          }
   }
   public static void main(String args[])
   {
     // float result = validate(10,2);
       float result = validate(5,0);
       System.out.println(result);

  }
}
```

Example 7

```
public class E_ThrowsDemo
{

     /*
     public static void main(String args[]) throws
ArithmeticException
      {
              int a = 10;
             int b = 0;

             int result = a / b;

             System.out.println(result);
```

```
      }
*/

    //loads the MySQL JDBC driver class
     //Class.forName("com.mysql.cj.jdbc.Driver");
    //example1
   public static void main(String args[]) throws
ClassNotFoundException
   {

              Class classlldr =
Class.forName("java.lang.Intege");
              System.out.println("Hello ");
   }



    /*
   public static void main(String args[]) //throws
ClassNotFoundException
   {
         try
         {
              Class classlldr =
Class.forName("java.lang.Intege");
         }
         catch(ClassNotFoundException e)
         {
             System.out.println("There is some pbm in
loading the class");
         }
     }
   */


    //example 2
    /*
    public static void main(String[] args)
    {
         try
         {
```

```
            Thread.sleep(10000); //Hello Geeks will  be
executed after 10000 milliseconds
            System.out.println("Hello Geeks");
        }
        catch(InterruptedException interruptedException)
        {
            System.out.println(interruptedException);
        }
    }
    */

    //example 3
    /*
    public static void main(String[] args)//throws
InterruptedException
    {
        Thread.sleep(100);
        System.out.println("Hello Geeks");
    }
     */
}
```